

MEMSYST STAGE 5 DEVELOPMENT MASTER PROMPT

Financial Infrastructure, Wallets, Revenue Distribution & Commercial Platform (FIP)

You are a Senior FinTech Architect, Enterprise Systems Architect, Multi-Tenant SaaS Architect, Revenue Systems Architect, Payment Infrastructure Architect, and Principal Engineer responsible for building Stage 5 of the MemSyst ecosystem.

Stages 1–4 have already established:

Stage 1

Business Operations Platform

- Leads
- CRM
- Tenant Onboarding
- Notifications
- Audit Infrastructure

Stage 2

Identity & Access Platform

- Authentication
- Authorization
- RBAC
- Permission Management
- Tenant Isolation

Stage 3

Tenant Management Platform

- Organizations
- Governance
- Executive Structures
- Branches
- Regions
- Tenant Configuration

Stage 4

Membership Platform

- Members
- Apprentices
- Member Lifecycle
- Renewals
- Membership Directory
- Membership Workflows

Stage 5 introduces the commercial and financial foundation of the entire MemSyst ecosystem.

This stage enables:

Membership Payments

Wallets

Revenue Sharing

Commissions

Messaging Billing

Financial Accountability

This is the stage where MemSyst begins generating platform revenue automatically.

STAGE 5 OBJECTIVE

Build the:

Financial Infrastructure Platform (FIP)

Purpose:

Provide a centralized financial operating system for:

MemSyst Platform

Tenant Organizations

National Structures

Regional Structures

Branches

Members

while ensuring:

Transparency
Traceability
Auditability
Revenue Distribution
Financial Governance

CRITICAL FINANCIAL PRINCIPLES

Principle 1

Wallet-Driven Architecture

Every financial transaction must pass through wallets.

No direct financial posting.

Principle 2

Ledger-Based Accounting

Every financial movement must generate:

Debit

Credit

Transaction Record

Audit Record

No balance should ever be manually edited.

Principle 3

Configurable Revenue Distribution

Revenue Distribution is configured by:

MemSyst Platform Admin

during tenant creation.

Only Platform Admin may update distribution models.

Tenant administrators cannot modify revenue distribution settings.

Principle 4

Multi-Level Financial Structure

The platform supports wallets for:

MemSyst

Tenant

National

Regional

Branch

Member

Principle 5

Immutable Financial Records

Financial transactions must never be deleted.

Corrections must occur through reversing entries.

STAGE 5 DELIVERABLES

Build:

Wallet Engine

Ledger Engine

Payment Engine

Revenue Distribution Engine

Commission Engine

Billing Engine

Financial Audit Engine

Financial Reporting Engine

MODULE 1

WALLET ENGINE

Purpose:

Provide digital wallets throughout the ecosystem.

Wallet Types

Platform Wallet

MemSyst Wallet

Tenant Wallet

Organization Wallet

National Wallet

National Office Wallet

Regional Wallet

Regional Wallet

Branch Wallet

Branch Wallet

Member Wallet

Member Wallet

Wallet Record

Wallet ID

Wallet Type

Owner ID

Tenant ID

Current Balance

Currency

Status

Created Date

Wallet Status

Active
Inactive
Suspended
Closed

MODULE 2

LEDGER ENGINE

Purpose:

Provide complete financial accountability.

Ledger Entry

Every transaction creates:

Transaction ID
Reference Number
Debit Wallet
Credit Wallet
Amount
Description
Timestamp

Rules

Never update balances directly.

All balances derive from ledger activity.

Ledger Features

Posting
Reversal
Reconciliation
Balance Calculation

MODULE 3

PAYMENT ENGINE

Purpose:

Manage collections.

Payment Types

Membership Registration

Membership Renewal

Apprentice Registration

Reprint Fees

Certificate Fees

Custom Tenant Fees

Payment Channels

Design for:

Mobile Money

Card Payments

Bank Transfer

Wallet Transfer

Payment Status

Pending

Processing

Successful

Failed

Cancelled

Refunded

MODULE 4

REVENUE DISTRIBUTION ENGINE

Purpose:

Automatically distribute collected revenue.

Distribution Model

Configured by:

Platform Admin

during tenant creation.

Example

Member Renewal Fee = GHS 100

Branch = 20%

Regional = 20%

National = 40%

Tenant Reserve = 10%

MemSyst = 10%

Requirements

Distribution percentages:

Configurable Per Tenant

Version Controlled

Auditable

Governance

Only:

MemSyst Super Admin

may update revenue distribution configurations.

MODULE 5

COMMISSION ENGINE

Purpose:

Manage platform earnings.

Commission Sources

Transaction Commissions
Messaging Markups
Future Service Fees

Commission Rules

Configurable.

No hardcoded percentages.

Tracking

Store:

Commission Source

Amount

Date

Related Transaction

MODULE 6

BILLING ENGINE

Purpose:

Manage financial obligations.

Bill Types

Membership Fees

Renewal Fees

Reprint Fees

Certificate Fees

Custom Fees

Billing Status

Draft

Pending

Due

Paid

Partially Paid
Cancelled

MODULE 7

MESSAGING BILLING ENGINE

Purpose:

Support centralized communication billing.

Existing Business Rule

MemSyst centrally manages:

SMS

Email

Push Notifications

In-App Notifications

Billing Workflow

Tenant Sends Message

↓

System Calculates Cost

↓

Tenant Wallet Checked

↓

If Balance Available:

Send Message

If Balance Insufficient:

Message Status = Pending

Message Status

Pending Funds

Queued

Processing

Sent

Failed

Track

Message Volume

Cost

Markup

Channel

Tenant

MODULE 8**FINANCIAL TRANSACTION MANAGEMENT**

Purpose:

Provide complete transaction visibility.

Transaction Record

Transaction ID

Reference Number

Source Wallet

Destination Wallet

Amount

Fee

Commission

Net Amount

Status

Timestamp

Filters

Tenant

Wallet

Transaction Type

Status

Date

MODULE 9

MEMBER FINANCIAL HISTORY

Purpose:

Provide member financial visibility.

Track

Registrations

Renewals

Reprints

Certificates

Other Charges

Features

View History

Download Receipt

Transaction Details

MODULE 10

FINANCIAL DASHBOARDS

Purpose:

Provide financial visibility.

Platform Dashboard

Display:

Revenue

Commissions

Messaging Revenue

Active Wallets

Transaction Volume

Tenant Dashboard

Display:

Collections
Revenue Distribution
Outstanding Bills
Wallet Balance

Branch Dashboard

Display:

Collections
Wallet Balance
Recent Transactions

MODULE 11

RECEIPT MANAGEMENT

Purpose:

Provide financial documentation.

Receipt Record

Receipt Number
Transaction ID
Amount
Date
Payer
Payment Method

Features

Generate
Download
Print
Verify

MODULE 12

REFUND MANAGEMENT

Purpose:

Manage financial corrections.

Refund Workflow

Request

↓

Review

↓

Approval

↓

Refund

Audit Requirements

Track:

Requester

Approver

Reason

Amount

MODULE 13

FINANCIAL REPORTING ENGINE

Purpose:

Provide financial reports.

Reports

Collections Report

Revenue Distribution Report

Commission Report

Wallet Report

Messaging Revenue Report

Outstanding Bills Report

Export Formats

PDF
Excel
CSV

MODULE 14

FINANCIAL AUDIT ENGINE

Purpose:

Provide financial accountability.

Audit Events

Payment Created
Payment Completed
Revenue Distributed
Wallet Credited
Wallet Debited
Refund Approved
Commission Applied

Audit Record

Actor
Action
Transaction
Amount
Date
Time
Before
After

MODULE 15

FINANCIAL SETTINGS

Purpose:

Centralized financial governance.

Settings Categories

Payment Settings
Commission Settings
Revenue Distribution Settings
Messaging Billing Settings
Currency Settings

Governance

Revenue Distribution Settings:
Platform Admin Only

MODULE 16

MEMBER SELF-SERVICE FINANCE

Purpose:
Allow members to manage financial activities.

Features

View Fees
Pay Fees
View Wallet
View Receipts
View History

MODULE 17

BRANCH & REGION FINANCIAL MANAGEMENT

Purpose:
Support decentralized financial operations.

Features

View Wallet
View Revenue
View Transactions
View Distributions

Structure

Support:

National
Regional
Branch

wallet ownership.

DATABASE DESIGN REQUIREMENTS

Before implementation:

Design complete entities for:

Wallets
Ledger Entries
Transactions
Bills
Invoices
Receipts
Revenue Models
Revenue Distributions
Commissions
Messaging Charges
Refunds
Financial Settings
Financial Audits

Define:

Relationships
Indexes
Validation Rules
Posting Rules
Reconciliation Rules

SECURITY REQUIREMENTS

Integrate with Stage 2.

Enforce:

RBAC

Tenant Isolation

Financial Permissions

Approval Permissions

Audit Logging

Sensitive Operations

Require elevated permissions for:

Revenue Model Changes

Refund Approvals

Manual Adjustments

Wallet Suspension

FUTURE READINESS REQUIREMENTS

Stage 5 must prepare for:

Marketplace Payments

Training Payments

Certificate Payments

AI Billing

Subscription Billing

Advanced Financial Analytics

Do not build them yet.

Design all financial relationships now.

ARCHITECTURE REQUIREMENTS

Use:

Double Entry Principles

Ledger-Based Accounting

Event-Driven Transactions

Feature-Based Architecture
Configuration-Driven Financial Rules

Avoid:

Direct Balance Editing
Hardcoded Commissions
Hardcoded Revenue Distribution
Deletable Transactions

All financial activity must be traceable.

SUCCESS CRITERIA

Stage 5 is complete only when:

- ✓ Wallets exist for all supported entities
- ✓ Ledger accounting is operational
- ✓ Payments can be processed
- ✓ Revenue distribution is automated
- ✓ Commission tracking is operational
- ✓ Messaging billing is operational
- ✓ Receipts are generated
- ✓ Refund workflows exist
- ✓ Financial reports are available
- ✓ Financial audits are immutable
- ✓ Branch and region financial structures are supported
- ✓ Platform monetization is operational
- ✓ Architecture is fully prepared for Stage 6 (Communication & Engagement Platform) and future marketplace/training financial integrations

Build this stage as the permanent financial backbone of the MemSyst ecosystem. Every future commercial activity, payment, commission, revenue share, and financial report must depend on the infrastructure established here.